

# Rajarshi Ray

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## EDUCATION

### Stony Brook University (SUNY)

Aug 2024 – Dec 2025

*M.S., Computer Science • GPA: 3.9/4.0 • Coursework: Computer Vision, NLP, AI, Data Science*

*New York, USA*

### Birla Institute of Technology and Science (BITS), Pilani

2020 – Jul 2024

*B.E., Computer Science • GPA: 8/10 • Teaching Assistant, Object-Oriented Programming*

*India*

## RESEARCH EXPERIENCE

### IBM × IBM Research

Feb 2026 – Present

*Staff Engineer, Artificial Intelligence*

*Poughkeepsie, NY*

- **InspectionBench (with IBM Research):** perception-grounded benchmark for embodied industrial anomaly verification, evaluating 4 agent architectures across 4,000+ real-world asset-inspection scenarios; manuscript targeting ICLR 2027.
- **Patent-pending** agentic AI system generating 100K+ persona-driven synthetic financial transactions per run for large-scale workload testing on IBM z17 LinuxONE, fully on-prem for enterprise security.
- Operate a local AI research playground on IBM GPU infrastructure: 10 self-hosted LLMs serving 20–25 engineers.

### Stony Brook University, VAI-GM Lab

Aug 2024 – Feb 2026

*Graduate Researcher, advised by Prof. Klaus Mueller*

*New York, USA*

- **NTIRE 2025 Super-Resolution (×4) Challenge:** led Team VAI-GM; designed MDRCT (Multi-step Dense Residual Connected Transformer) extending the DRCT baseline; achieved 30.86 dB PSNR / 0.8942 SSIM, ranked 18th globally.
- **3D lung CT synthesis:** generated high-fidelity synthetic CT volumes via VQ-VAE latent encoding and guided diffusion; deployed a Dockerized Orthanc DICOM server for secure image storage.
- **All-in-one image restoration:** curated 80,000+ grid images (rain, haze, noise, snow) and trained a single transformer to restore multiple degradations.

## PUBLICATIONS & TECHNICAL WRITING

**Ray, R.**, et al. “InspectionBench: Embodied Industrial Anomaly Verification with Perception-Grounded Physical AI Evaluation.” *In preparation* (with IBM Research), targeting ICLR 2027.

Chen, Z., **Ray, R.**, Tomar, S.S., Mueller, K., et al. (2025). “NTIRE 2025 Challenge on Image Super-Resolution (×4): Methods and Results.” *CVPR Workshops (NTIRE)*. [arXiv:2504.14582](https://arxiv.org/abs/2504.14582).

**Ray, R.** (2026). “From Prompt to Pipeline: AI-Powered Transaction Generation and Workload Testing on IBM Z LinuxONE.” *IBM Z & LinuxONE Community*.

**Ray, R.** (2026). “Synthetic Transaction Generation Powered by AI Agents.” *IBM Z & LinuxONE Community*.

**Ray, R.** (2025). “Porting and Deployment of the AWS Retail Store Application on IBM s390x using OpenShift.” *IBM Z & LinuxONE Community*.

## INDUSTRY & ENTREPRENEURIAL EXPERIENCE

### SceneSenseAI

2026 – Present

*Founder (part-time)*

*New York, USA*

- Building personalized AI teaching assistants over lecture video libraries: learners ask questions in natural language and get answers cited to exact timestamps in the source videos.
- Designed an instructor-facing analytics engine surfacing learner confusion points and content gaps per module; validating with course creators and university instructors through a US pilot program.

### IUDX, Indian Institute of Science (IISc)

Jan 2024 – Jul 2024

*Research and Development Intern*

*Bangalore, India*

- Developed a secure inference workflow for pneumonia classification on AMD SEV-SNP, a 20% performance gain over comparable stacks (Intel SGX, AWS Nitro) while preserving data privacy and code integrity.
- Designed a confidential medical image analysis pipeline integrating TPM-based storage with remote inference (TensorFlow, Keras, C++, Docker).

## TECHNICAL SKILLS

**AI / ML:** PyTorch, TensorFlow, Diffusion Models, VAEs, GANs, Transformers, LLMs, Agentic AI, RAG, Hugging Face, OpenCV, YOLO

**Systems & Languages:** Docker, Podman, OpenShift/Tekton, Kubernetes, Linux (x86/s390x), JMeter, Confidential Computing (SEV-SNP, TPM), Orthanc/DICOM • Python, C/C++, Java, Kotlin, SQL, Bash